ZHANHAO HU

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EDUCATION

Tsinghua University, Beijing *2017 - 2023*
Ph.D., Computer Science and Technology
Advisor: Bo Zhang and Xiaolin Hu
Dissertation: The Practicality of Physical Adversarial Examples for Deep Learning Models.
Research: adversarial robustness, privacy, and brain-inspired algorithms in deep learning.

Tsinghua University, Beijing *2013 - 2017*
B.S., Mathematics and Physics
Advisor: Xiaolin Hu
Dissertation: STDP-based Learning for Spiking Neural Networks

RESEARCH INTERESTS

My research develops principled defenses and red-teaming methods for secure and robust AI systems, including large language models (LLMs), agentic systems, and multimodal perception systems. My current interests include alignment-preserving finetuning, prompt injection defenses, secure agentic system design, and multimodal adversarial robustness.

RESEARCH EXPERIENCE

University of California, Berkeley January 2024 - Present
Postdoctoral Researcher *Berkeley, California*

- Advised by Prof. David Wagner, affiliated with the Department of Electrical Engineering and Computer Science (EECS) and the Institute for Data Science (BIDS)
- Research focuses on LLM security and safety, including model-level and agentic system-level attacks and defenses.
- Led 6+ projects on red-teaming, detection, and defense across threat models in LLM and embodied systems.
- Collaborated with students and faculty on LLM security, agentic system security, and multimodal robustness.

Tsinghua University July 2023 - December 2023
Researcher *Beijing, China*

- Worked with Prof. Xiaolin Hu at the Tsinghua Laboratory of Brain and Intelligence (THBI).
- Research focused on physical adversarial robustness for computer vision models.

PUBLICATIONS

(Sorted by time; * for equal contribution)

Under review* & *Preprint

1. **Zhanhao Hu**, Dennis Jacob, Xiao Huang, Zhaorun Chen, Bo Li, and David Wagner (2026) Twin Agent: Context Residual Compression for Privilege Separated Agents (manuscript in preparation)
2. **Zhanhao Hu***, Xiao Huang*, Patrick Mendoza, Emad A. Alghamdi, Basel Alomair, Raluca Ada Popa, and David Wagner (2026). GradShield: Alignment Preserving Finetuning (preprint)
3. Zi Yin, Peilin Chai, Siyuan Huang, **Zhanhao Hu** (2026). Test-time Adversarial Takeover: A Real-time Hijacking Interface against Robotic Diffusion Policies (preprint)
4. Jesson Wang, **Zhanhao Hu** (2026). Safety Offset Learning: When Benign Fine-Tuning Misaligns LLMs (under review)
5. Dennis Jacob, Emad Alghamdi*, **Zhanhao Hu***, Basel Alomair, and David Wagner (2025). Better Privilege Separation for Agents by Restricting Data Types (preprint)
6. Qiongxiu Li, Lixia Luo, Agnese Gini, Changlong Ji, **Zhanhao Hu**, Xiao Li, Chengfang Fang, Jie Shi, Xiaolin Hu (2024). Perfect Gradient Inversion in Federated Learning: A New Paradigm from the Hidden Subset Sum Problem (preprint)

Published

7. Jesson Wang*, **Zhanhao Hu***, and David Wagner (2026). JULI: Jailbreak Large Language Models by Self-Introspection (ICLR).
8. Xiaopei Zhu, Guanning Zeng, **Zhanhao Hu**, Jun Zhu, and Xiaolin Hu (2026). Physical Adversarial Clothing Evades Visible-Thermal Detectors via Non-Overlapping RGB-T Pattern (CVPR)
9. Julien Piet, Xiao Huang, Dennis Jacob, Annabella Chow, Maha Alrashed, Geng Zhao, **Zhanhao Hu**, Chawin Sitawarin, Basel Alomair, and David Wagner (2025). Jailbreaksovertime: Detecting jailbreak attacks under distribution shift (AISec)
10. Dennis Jacob, Hend Alzahrani, **Zhanhao Hu**, Basel Alomair, and David Wagner (2025). Prompt-Shield: Deployable Detection for Prompt Injection Attacks (CODASPY)
11. Xiaopei Zhu, Siyuan Huang, **Zhanhao Hu**, Jianmin Li, Jun Zhu, Xiaolin Hu (2025). Physical Adversarial Examples for Person Detectors in Thermal Images Based on 3D Modeling. (TPAMI)
12. **Zhanhao Hu**, Julien Piet, Geng Zhao, Jiantao Jiao, and David Wagner (2024). Toxicity Detection for Free. Advances in Neural Information Processing Systems (NeurIPS Spotlight)
13. Zhi Cheng, **Zhanhao Hu**, Yuqiu Liu, Jianmin Li, Hang Su, Xiaolin Hu (2024). Full-Distance Evasion of Pedestrian Detectors in the Physical World. Advances in Neural Information Processing Systems (NeurIPS)
14. Xiao Li, Wei Zhang, Yining Liu, **Zhanhao Hu**, Bo Zhang, Xiaolin Hu (2024). Language-Driven Anchors for Zero-Shot Adversarial Robustness. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
15. Xiao Li, Qiongxiu Li, **Zhanhao Hu**, Xiaolin Hu (2024). On the Privacy Effect of Data Enhancement via the Lens of Memorization. IEEE Transactions on Information Forensics and Security (IEEE TIFS).

16. Xiaopei Zhu, Yuqiu Liu, **Zhanhao Hu**, Jianmin Li, and Xiaolin Hu (2024) Infrared Adversarial Car Stickers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
17. Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Xiaolin Hu (2023). Hiding from Infrared Detectors in Real World with Adversarial Clothes. Applied Intelligence.
18. **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Hui Zhang, Bo Zhang, Xiaolin Hu (2023). Physically Realizable Natural-Looking Clothing Textures Evade Person Detectors via 3D Modeling. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
19. Tong Wang, Xiaohui Kuang, Qianjin Du, **Zhanhao Hu**, Huan Deng, Gang Zhao. (2023). Driving into Danger: Adversarial Patch Attack on End-To-End Autonomous Driving Systems Using Deep Learning. In 2023 IEEE Symposium on Computers and Communications (ISCC)
20. **Zhanhao Hu**, Siyuan Huang, Xiaopei Zhu, Fuchun Sun, Bo Zhang, Xiaolin Hu (2022). Adversarial texture for fooling person detectors in the physical world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Oral).
21. Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Xiaolin Hu (2022). Infrared invisible clothing: Hiding from infrared detectors at multiple angles in real world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Oral).
22. **Zhanhao Hu**, Jun Zhu, Bo Zhang, Xiaolin Hu (2022). Amplification Trojan network: Attack deep neural networks by amplifying their inherent weakness. Neurocomputing, 505, 142-153.
23. **Zhanhao Hu**, Tao Wang, Xiaolin Hu (2017). An STDP-based supervised learning algorithm for spiking neural networks. In Neural Information Processing: 24th International Conference (ICONIP).

AWARDS & HONORS

1. NeurIPS Spotlight (top 2%), ”Toxicity Detection for Free,” 2024
2. CVPR Oral Presentation (top 4%), ”Adversarial Texture for Fooling Person Detectors,” 2022
3. CVPR Oral Presentation (top 4%), ”Infrared Invisible Clothing,” 2022
4. Recognized as Outstanding Graduate of the Computer Science Department, 2023

TEACHING & MENTORSHIP

- *Teaching Assistant*, Neural and Cognitive Computation (No.80240642), Tsinghua University, Autumn 2018 & Autumn 2019. — Assisted with tutorials and grading; Delivered a guest lecture on deep learning tooling (PyTorch/TensorFlow).
- *Mentor*, UC Berkeley, 2024–Present — Mentored 3+ undergraduate students in LLM security and safety research.
- *Mentor*, Tsinghua University, 2017–2023 — Mentored 5+ undergraduate and junior Ph.D. students in adversarial example research.

PROFESSIONAL SERVICE

Reviewer

- Journal Reviewer: TIP, TPAMI, TNNLS, PR
- Conference Reviewer: NeurIPS, ICML, CVPR, ICLR, ICCV, ECCV, AAAI, EMNLP, ICIST, ICACI, ICICIP, ISNN

PATENTS

1. Xiaolin Hu, **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Three-dimensional modeling method and apparatus for flexible object, electronic device, and storage medium, WO2024250729A1, 2024-12-12
2. Xiaolin Hu, **Zhanhao Hu**, Wenda Chu, Xiaopei Zhu, Method and apparatus for generating camouflage pattern, electronic device, and storage medium, WO2024234714A1, 2024-11-21
3. Xiaolin Hu, Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Manufacturing method and using method of infrared camouflage pattern and infrared camouflage garment, CN114557507A, 2022-05-31
4. Xiaolin Hu, Xiaopei Zhu, **Zhanhao Hu**, Siyuan Huang, Jianmin Li, Countermeasure image generation method and apparatus, and target cover processing method, CN114550217A, 2022-05-27
5. Xiaolin Hu, **Zhanhao Hu**, Xiaopei Zhu, Siyuan Huang, Target detector safety testing method and device, electronic device and storage medium, CN113780280B, 2021-12-10
6. Xiaolin Hu, **Zhanhao Hu**, Xiaopei Zhu, Siyuan Huang, Target detector safety testing method and device, electronic device and storage medium, CN113780433B, 2021-12-10